

Lecture 7: Foreign Exchange

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Overview

This lecture introduces foreign exchange rates and exchange markets, helping you understand how exchange rates are determined and what accounts for their fluctuation over time.

Key insights:

- Exchange rates are the price of one currency in terms of another
- They make international trade possible by facilitating currency exchange
- Changes in exchange rates affect costs of goods and services across countries

The Road Ahead

1. Foreign Exchange Basics
2. Exchange Rates in Long Run
3. Exchange Rates in Short Run

Nominal Exchange Rate

Let E be the nominal exchange rate - the price of domestic currency in terms of foreign currency. Example: If dollar is domestic and pound is foreign, E denotes the price of a dollar in terms of pounds.

Exchange rates change every day:

1. **Nominal appreciation** ($\uparrow E$): Domestic currency becomes more expensive in terms of foreign currency
2. **Nominal depreciation** ($\downarrow E$): Domestic currency becomes less expensive in terms of foreign currency
3. **Fixed exchange rate systems**: Increases in E are called revaluations, decreases are called devaluations

Real Exchange Rate

The real exchange rate ϵ is the price of U.S. goods in terms of British goods.



Real exchange rates also move over time:

1. **Real appreciation** ($\uparrow \epsilon$): An increase in the relative price of domestic goods in terms of foreign goods
2. **Real depreciation** ($\downarrow \epsilon$): A decrease in the relative price of domestic goods in terms of foreign goods

Key insight: Nominal and real exchange rates can move in opposite directions, but movements in E typically dominate, so ϵ inherits large fluctuations from E .

Uncovered Interest Parity Condition

Consider the choice between U.S. and U.K. one-year bonds from a U.S. investor's viewpoint.

- U.S. bonds return: $(1 + i_t)$ dollars per dollar invested
- U.K. bonds return: $E_t(1 + i_t^*)/E_{t+1}^e$ dollars per dollar invested

Arbitrage condition:

$$1 + i_t = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right)$$

Approximation of Interest Parity

$$\underbrace{i_t}_{\text{domestic rate}} \approx \underbrace{i_t^*}_{\text{foreign rate}} - \underbrace{\frac{E_{t+1}^e - E_t}{E_t}}_{\text{expected appreciation of domestic currency}}$$

Factors ignored: Transaction costs and exchange rate risk.

Bond example ($i_t = 2\%$ U.S., $i_t^* = 5\%$ U.K.):

- U.S. bonds attractive if pound depreciates $> 3\%$
- U.K. bonds attractive if pound depreciates $< 3\%$

Under fixed rates: If $E_{t+1}^e = E_t$, then $i_t \approx i_t^*$

Exchange Rates in Long Run

We now examine how nominal exchange rates are determined in the long run (e.g., decades).

Two fundamental concepts:

1. The law of one price
2. Purchasing power parity (PPP)

Both are based on the concept of **no arbitrage**: identical products should sell for the same price.

Law of One Price

The law of one price states that identical products should sell for the same price when measured in the same currency.

Example: TV selling in Detroit and Windsor (Canada)

$$\$P_{\text{Windsor}} = \$P_{\text{Detroit}} \times E$$

where E is the dollar-Canadian dollar exchange rate.

Limitations: This law can fail easily in the short run due to:

- Transportation costs
- Tariffs and trade barriers
- Different local taxes

Purchasing Power Parity

Purchasing power parity (PPP) extends the law of one price to a basket of goods and services.

$$1 = \frac{\text{dollar price of one U.S. basket}}{\text{dollar price of one U.K. basket}} = \frac{P}{P^*/E} = \epsilon$$

Rearranging the PPP condition:

$$\underbrace{E}_{\text{pounds per dollar}} = \frac{\text{pound price of one basket in U.K.}}{\text{dollar price of one basket in U.S.}} = \frac{P^*}{P}$$

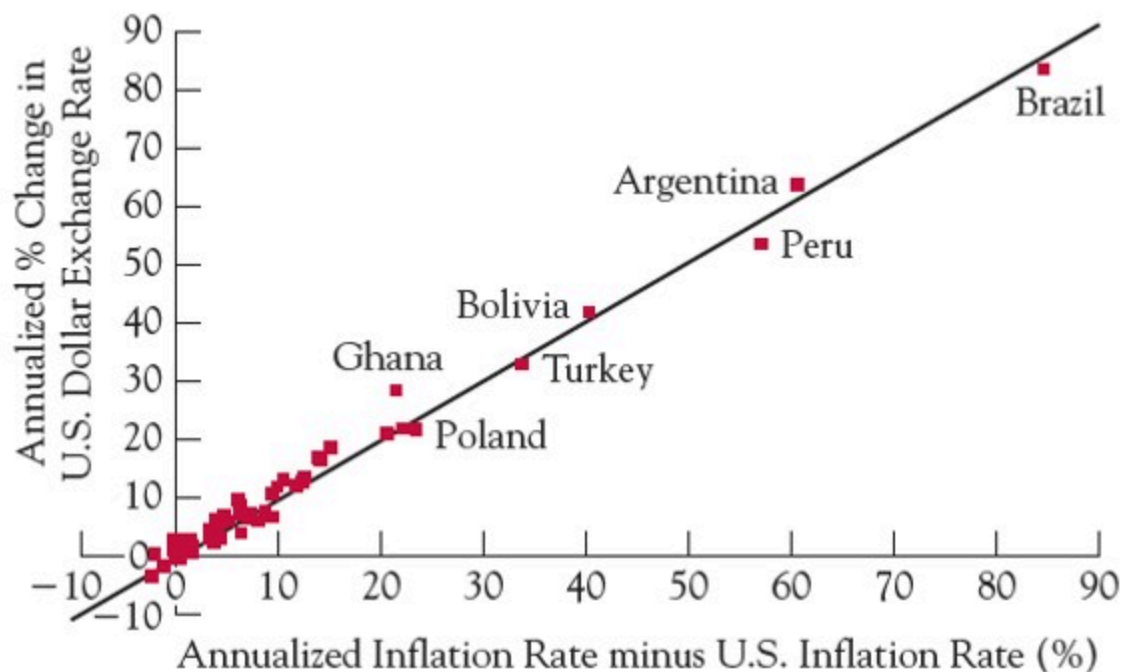
Key insight: Changes in exchange rates are tied to inflation differences between countries.

Exchange Rate Determination

The currency of a country with high inflation will depreciate.

Figure 10.4

Exchange Rate Movements and Inflation Differentials, 1980–2010



Supply and Demand for Dollars

Supply of dollars:

- Americans want to exchange dollars for foreign currencies to:
 - i. Purchase goods and services produced abroad
 - ii. Invest in foreign assets
- Higher exchange rate → more dollars supplied

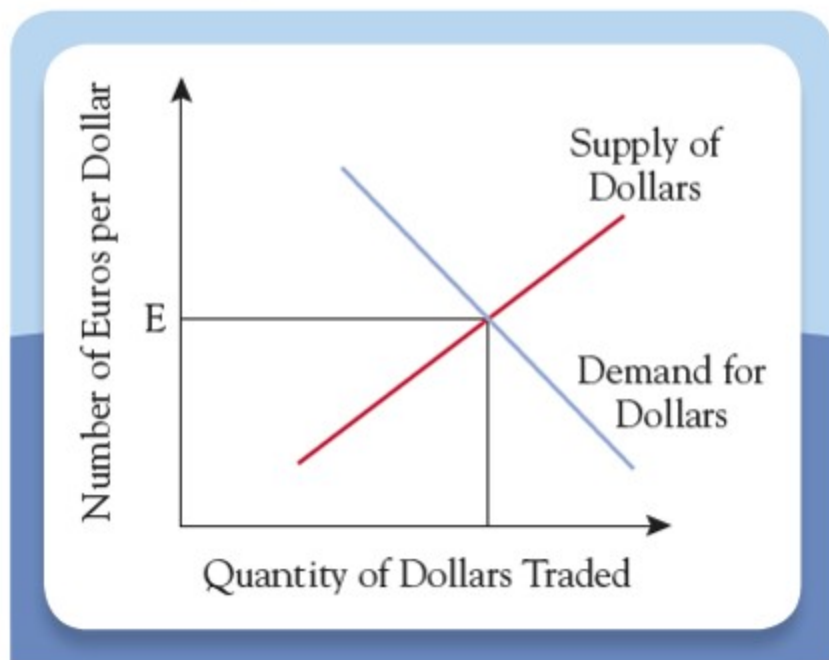
Demand for dollars:

- Foreigners want dollars to purchase U.S. goods, services, and assets
- Lower exchange rate → more dollars demanded

Dollar Market Equilibrium

Figure 10.5

The Dollar–Euro Market



Shifts in Dollar Supply and Demand

Supply increases:

1. Americans prefer foreign goods
2. Higher foreign interest rates
3. Increase in American wealth
4. Lower foreign investment risk
5. Expected dollar depreciation

Demand increases:

1. Foreigners prefer U.S. goods
2. Higher U.S. interest rates
3. Increase in foreign wealth
4. Lower U.S. investment risk
5. Expected dollar appreciation